

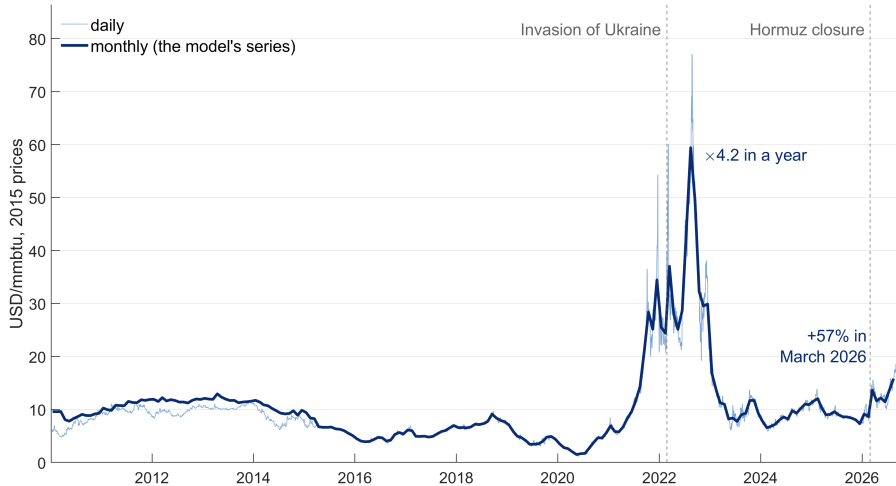
Revisiting Fundamentals of the European Gas Market

The Role of Supply Substitution

Jorge Arenas Amorós

Universidad de Alicante · Autumn 2026

European gas prices



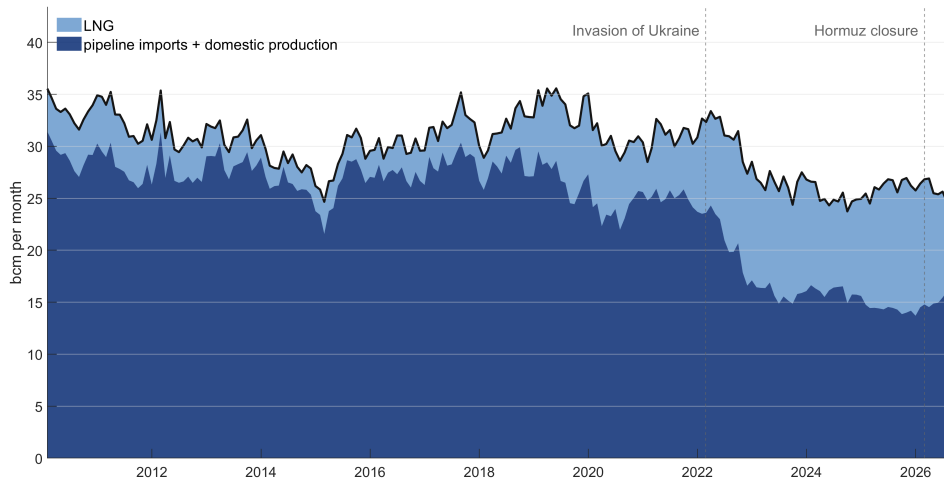
HICP-deflated. Daily: TTF front-month (LSEG). Monthly: World Bank (border price before April 2015).

Total gas supply



22 EU members (JODI), seasonally adjusted. Total = pipeline + domestic production + LNG.

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Research questions

1. What drives European gas prices?
2. Does it matter where Europe's gas supply comes from?
3. What can we learn from 2022 to make policy recommendations for 2026?

This paper

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- Separate pipeline gas from LNG and identify the elasticities of each channel
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- Decompose the main drivers of the price
 - ▶ *Storage demand is the main driver of the price.*
 - ▶ *Expectations effect is immediate and partly fades.*
 - ▶ *Supply shocks build more slowly and persist.*

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- Decompose the main drivers of the price
 - ▶ *Storage demand is the main driver of the price.*
 - ▶ *Expectations effect is immediate and partly fades.*
 - ▶ *Supply shocks build more slowly and persist.*
- Interpret the 2022 and 2026 crises and quantify policy channels with counterfactuals
 - ▶ *Saving gas contained the 2022 price more than extra LNG.*

Data

Six monthly variables, February 2014 to July 2026 (150 observations), log growth rates:

- q^{pip} : pipeline net imports plus domestic production
- q^{LNG} : LNG net imports
- s : gas in storage
- c : observed inland gas consumption
- p : real TTF price
- ip : euro-area manufacturing

Sources: JODI (supply and storage, 22 EU members), Eurostat (consumption, manufacturing, HICP), World Bank (price). Quantities seasonally adjusted (X-13). The sample includes the first five months of the Hormuz crisis.

The model

Structural VAR, $\mathbf{A}y_t = \mathbf{B}x_{t-1} + u_t$; its contemporaneous part:

Pipeline supply $q_t^{pip} = \gamma_{qp} p_t + u_t$ $\gamma_{qp} > 0$

LNG supply $q_t^{LNG} = \beta_{qp} p_t + u_t$ $\beta_{qp} > 0$

Priors and posteriors

Daily evidence for the storage priors

Posterior formulas

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Storage	$s_t = \psi_1 q_t^{pip} + \psi_2 p_t + \psi_3 q_t^{LNG} + u_t$	no sign restrictions

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Storage $s_t = \psi_1 q_t^{pip} + \psi_2 p_t + \psi_3 q_t^{LNG} + u_t$ no sign restrictions

Market balance $c_t = (1 - w_L) q_t^{pip} + w_L q_t^{LNG} - w_S s_t + u_t$ $0 < w_L < 1, w_S > 0$

Priors on the elasticities from the literature; storage priors measured from daily data.

Priors and posteriors

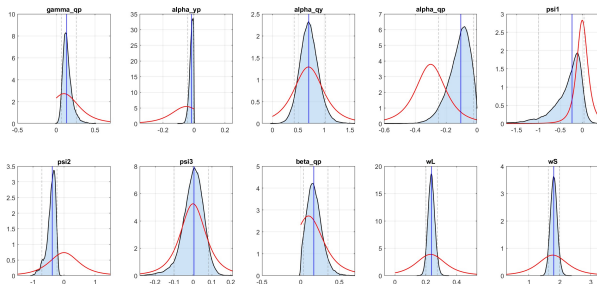
Daily evidence for the storage priors

Posterior formulas

Posterior elasticities

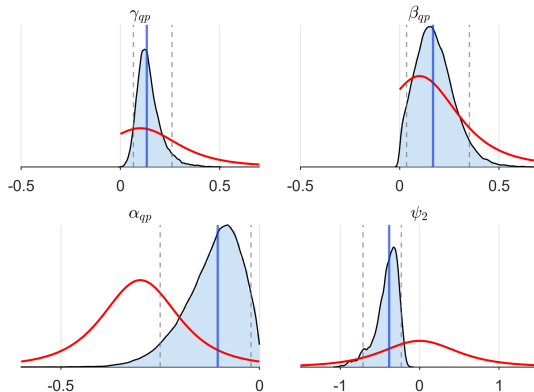
Posterior median [90% interval]

- Pipeline $\gamma_{qp} = 0.13$ [0.07, 0.26]
- LNG $\beta_{qp} = 0.17$ [0.03, 0.35]
 - ▶ $\Pr(\beta_{qp} > \gamma_{qp}) = 0.60$
- Demand $\alpha_{qp} = -0.10$ [-0.25, -0.02]
 - ▶ a 10% price rise cuts consumption by 1%
- Storage $\psi_2 = -0.38$ [-0.71, -0.23]
 - ▶ higher prices release gas



*Prior (red line) and posterior (histogram, kernel density);
blue line = posterior median, dashed = 5% and 95%
quantiles.*

Posterior elasticities

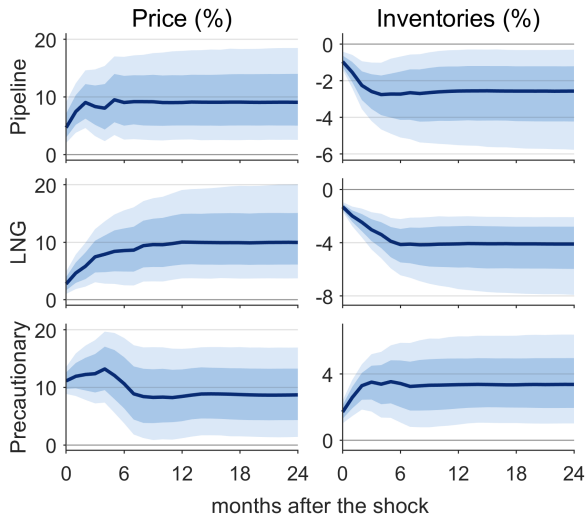


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Impulse responses: slow supply, fast expectations

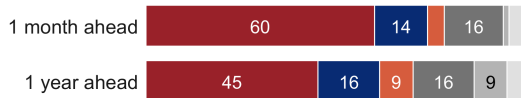


- *Supply shocks build slowly and persist: pipeline peaks at 5 months, LNG at 12.*
- *The precautionary shock hits hardest (+11%); a third of it fades within two years.*
- *Inventories rise after a precautionary shock and fall after supply shocks.*

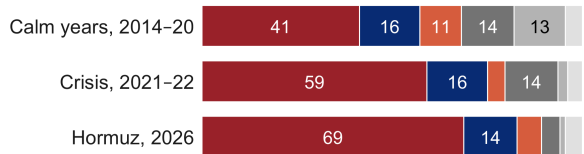
Cumulative responses (%) to a one-s.d. shock; median with 68% and 90% bands. Pipeline and LNG shocks shown as adverse.

Variance decomposition: storage demand dominates

Forecast error variance



Realised variance, by period

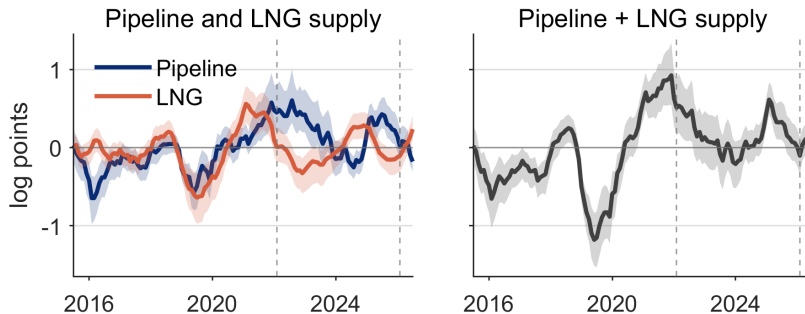


Precautionary Pipeline LNG
Flow demand Aggregate demand Residual

- Storage demand explains close to half of the price variance (45% at one year).
- It dominates in calm years too (41%) and more under stress (59% in 2021–22, 69% in 2026).
- Supply shocks explain far less: pipeline 16%, LNG 9%.

Shares (%) of the variance of the real price, posterior means. Bottom: realised variance within each period (2026: March to July).

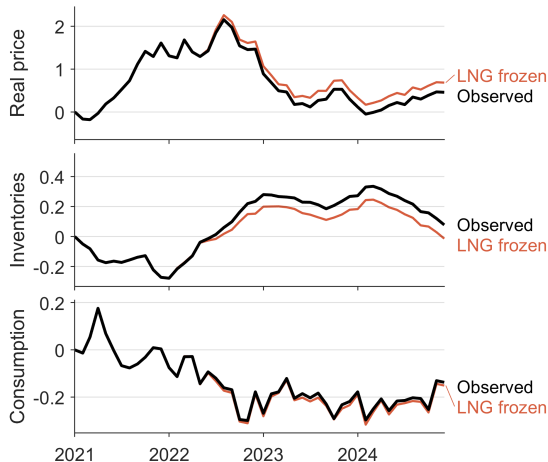
Historical decomposition



Contribution to the 12-month change of the log real price; median and 68% bands.

- ▶ *LNG led the supply push in 2021, pipeline gas in 2022: summed, the switch disappears.*
- ▶ *In 2023 and in summer 2026 they offset each other, so the sum stays near zero.*
- ▶ *About a fifth of their movement cancels out when they are added up.*

Three what-ifs for 2022–23



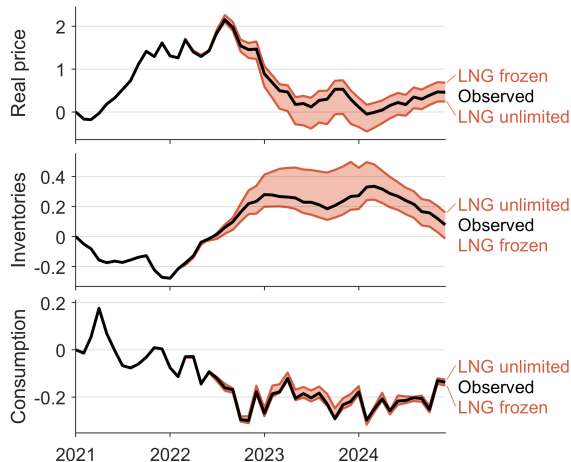
Log levels, January 2021 = 0.

- With LNG frozen, inventories would have ended 8% lower while consumption barely moved.

Gap vs observed, December 2023 (%)

	price	stocks	cons.
LNG frozen	+23	-8	-2

Three what-ifs for 2022–23



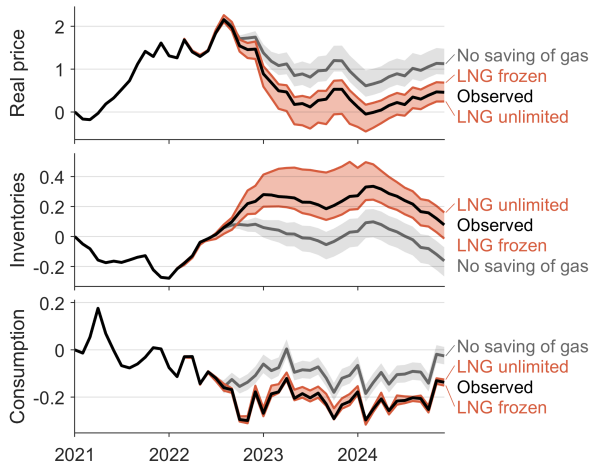
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- Between LNG frozen and LNG unlimited, the price would have been 23% higher or 43% lower.

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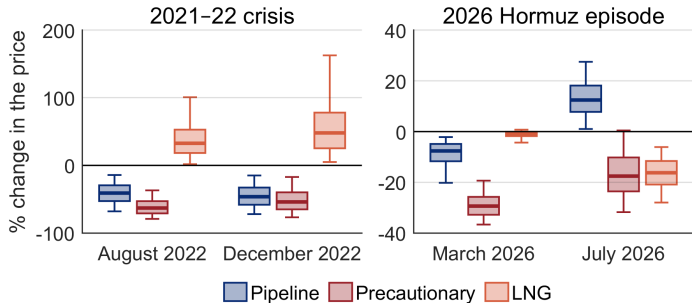
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- With LNG frozen, inventories would have ended 8% lower while consumption barely moved.
- Between LNG frozen and LNG unlimited, the price would have been 23% higher or 43% lower.
- Without the saving of gas, the price would have been about twice as high.

Gap vs observed, December 2023 (%)

	price	stocks	cons.
LNG frozen	+23	-8	-2
LNG unlimited	-43	+26	+3
No saving of gas	+99	-21	+12

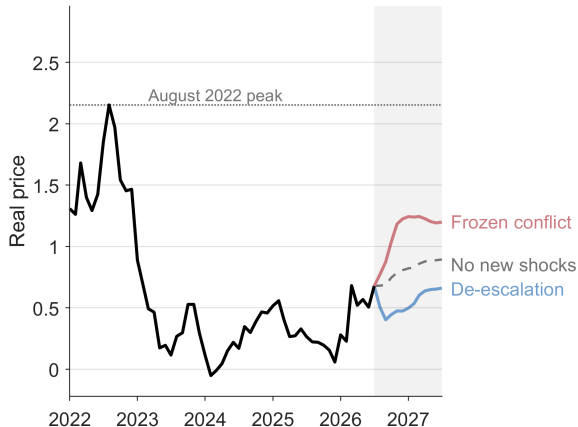
The two crises through the same lens



Price change if each channel's shocks are removed, from September 2021 and from March 2026; boxes: 25-75%, whiskers: 5-95%.

- *In both crises precautionary demand made the peak: without it the price would have been 63% lower in August 2022 and 29% lower in March 2026.*
- *Cargoes came later: LNG barely mattered at the March peak, but by July it weighed as much as precautionary demand (16% vs 18%).*

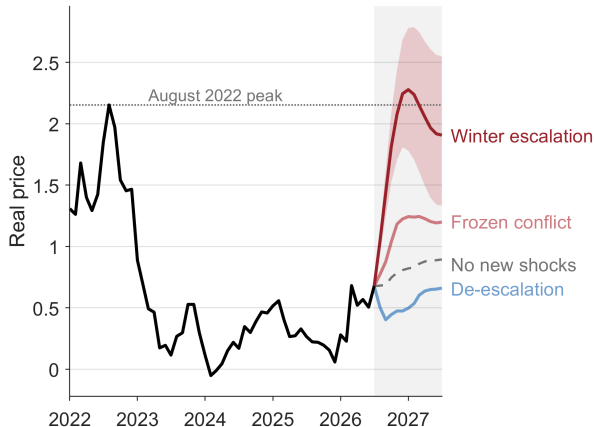
Hormuz: three forward scenarios (illustrative)



- A frozen conflict would keep the price about half above the no-new-shocks path this winter; de-escalation, about a quarter below.

Log level, January 2021 = 0. Shaded: scenarios calibrated by us, shocks in units of their March 2026 size; medians.

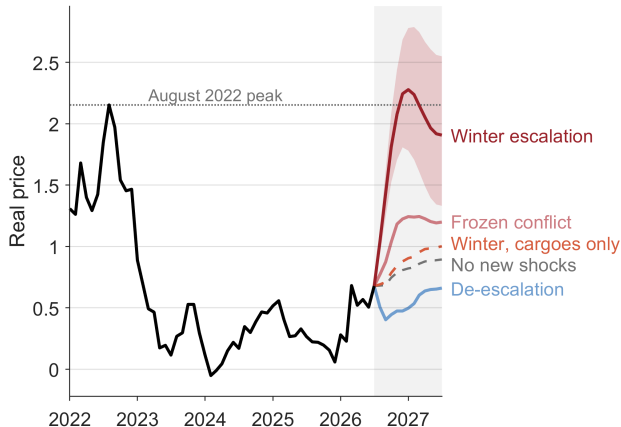
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- *A winter escalation would take the price back to around its 2022 peak this winter.*

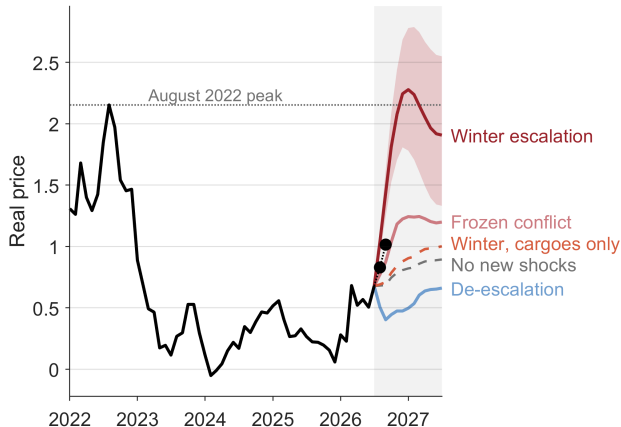
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- ▶ *At least three quarters of that rise would come from precautionary demand, not from the missing cargoes.*

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- A frozen conflict would keep the price about half above the no-new-shocks path this winter; de-escalation, about a quarter below.
- A winter escalation would take the price back to around its 2022 peak this winter.
- At least three quarters of that rise would come from precautionary demand, not from the missing cargoes.
- Since July the price has risen 40%, between frozen conflict and winter escalation.

Take-aways

1. Separating pipeline gas from LNG: the two price elasticities are **small and similar**, LNG somewhat the more elastic, but their roles in the crises were not the same, and an aggregate model loses the composition of supply where the adjustment took place.
2. The European gas price is set by **expectations operating through storage**: precautionary demand carries about half of its variance, in calm years and, above all, under stress.
3. Channels differ in **timing**: expectational shocks act at once and partly unwind, physical supply shocks build over months and stay. Which channel is at work tells you how long a price spike will last.
4. Policy, 2022–23: supply diversification and the saving of gas both mattered, but **not equally**; the saving of gas contained the price by more, and 2026 shows the cost of leaving expectations unanchored.

Robustness

Alternative supply data (IEA)

Thank you

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Appendix

Data

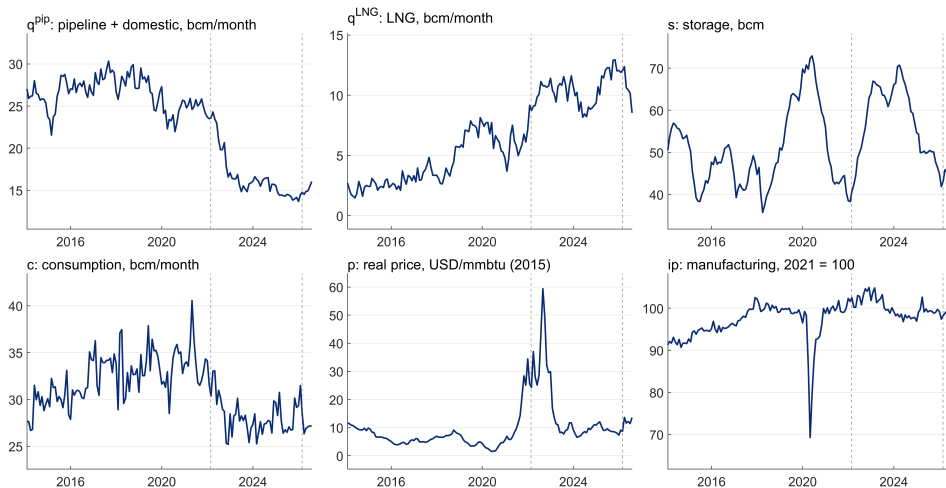
Identification and priors

Dynamics

Counterfactuals

Robustness

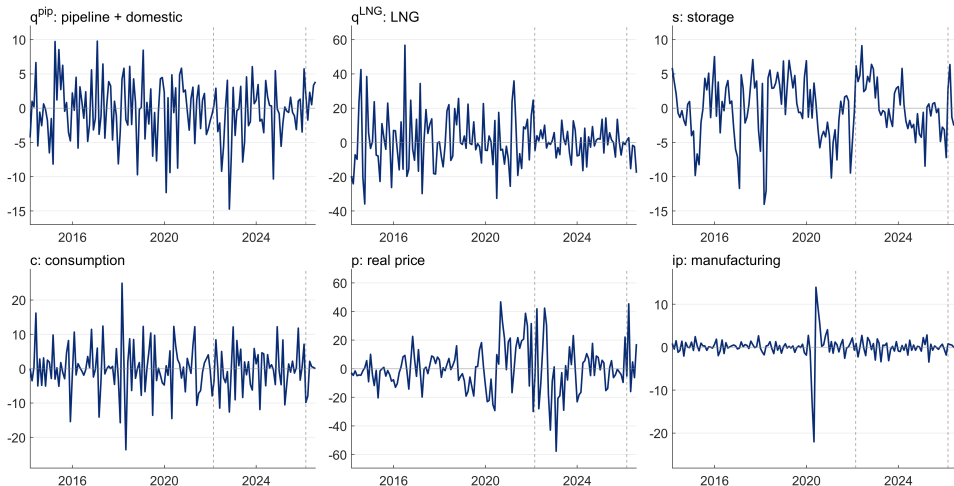
The six series



Levels; the model uses their monthly log growth rates. Quantities seasonally adjusted.

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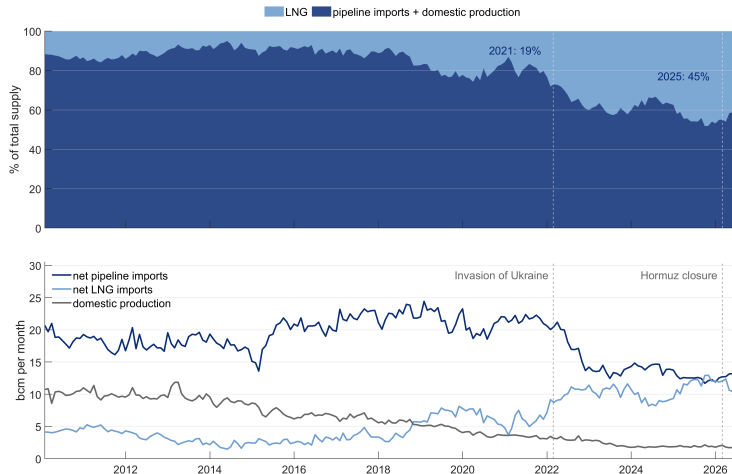
The six series in log growth rates



$100 \times \Delta \log$, February 2014 to July 2026 (150 observations): the variables of the VAR.

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Composition of supply: shares and components



Top: shares of total supply. Bottom: components, bcm per month. Seasonally adjusted, JODI.

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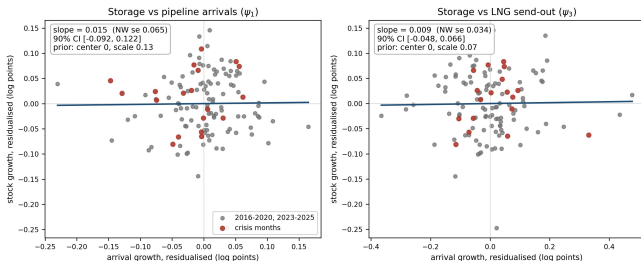
Priors and posteriors of the ten contemporaneous parameters

Parameter	Interpretation	Prior	Posterior		
		mode (sd)	5%	median	95%
γ_{qp}	pipeline supply elasticity	0.10 (0.20)	+0.065	+0.133	+0.259
α_{yp}	activity response to price	-0.05 (0.10)	-0.041	-0.015	-0.002
α_{qy}	demand income elasticity	0.70 (0.30)	+0.422	+0.706	+1.022
α_{qp}	demand price elasticity	-0.30 (0.10)	-0.249	-0.105	-0.022
ψ_1	storage: pipeline supply	0.00 (0.13)	-1.001	-0.236	+0.042
ψ_2	storage: price	0.00 (0.50)	-0.713	-0.382	-0.228
ψ_3	storage: LNG supply	0.00 (0.07)	-0.100	+0.004	+0.083
β_{qp}	LNG supply elasticity	0.10 (0.20)	+0.034	+0.167	+0.351
w_L	LNG share in supply	0.23 (0.10)	+0.201	+0.236	+0.274
w_S	stock/consumption ratio	1.77 (0.50)	+1.626	+1.804	+1.990

Prior mode and scale of the truncated Student- t ; posterior quantiles from a chain of 2 million draws (300,000 retained, thinned to 30,000; acceptance 0.27). The scales of ψ_1 and ψ_3 (0.13 and 0.07) are twice the standard errors of the daily-data regressions.

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Daily evidence behind the storage priors



Monthly EU stock growth against the growth of pipeline arrivals (left) and LNG send-out (right), both residualised on the other aggregate, the price, heating demand and month effects, 2016–2026. Crisis months in red.

Daily data from Gas Infrastructure Europe (storage, LNG terminals) and ENTSOG (pipeline entry points), aggregated to months.

Slopes: 0.015 (Newey–West s.e. 0.065) for pipeline, 0.009 (0.035) for LNG.

Priors on ψ_1, ψ_3 centred at zero with scales of twice the standard errors, 0.13 and 0.07.

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Posterior formulas (Baumeister and Hamilton, 2015)

Joint posterior

$$p(\mathbf{A}, \mathbf{B}, \mathbf{D} \mid Y_T) = p(\mathbf{A} \mid Y_T) p(\mathbf{D} \mid \mathbf{A}, Y_T) p(\mathbf{B} \mid \mathbf{A}, \mathbf{D}, Y_T)$$

Posterior of $\mathbf{D}$ (diagonal, inverse Gamma):

$$d_{ii}^{-1} \mid \mathbf{A}, Y_T \sim \Gamma(\kappa_i^*, \tau_i^*(\mathbf{A})), \quad \kappa_i^* = \kappa_i + \frac{T}{2}, \quad \tau_i^*(\mathbf{A}) = \tau_i + \xi_i^*(\mathbf{A})$$

Posterior of $\mathbf{B}$ (conditional normal):

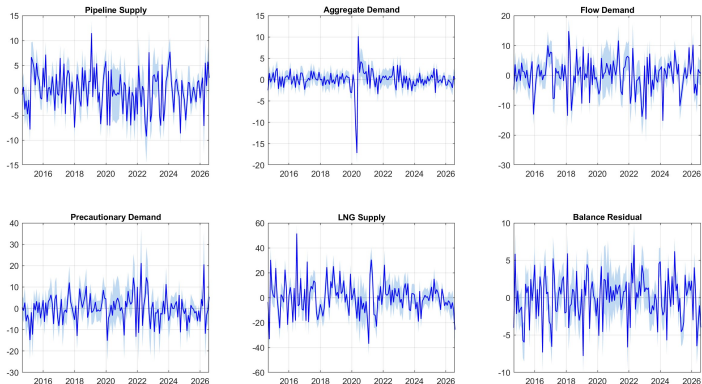
$$b_i \mid \mathbf{A}, \mathbf{D}, Y_T \sim \mathcal{N}(m_i^*, d_{ii} M_i^*), \quad m_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1} \tilde{X}_i' \tilde{Y}_i, \quad M_i^* = (\tilde{X}_i' \tilde{X}_i)^{-1}$$

Posterior of $\mathbf{A}$ (no closed form, Metropolis–Hastings):

$$p(\mathbf{A} \mid Y_T) = k_T p(\mathbf{A}) \left| \det(\mathbf{A} \hat{\Omega}_T \mathbf{A}') \right|^{-T/2} \prod_{i=1}^n \left(\frac{2\tau_i^*}{T} \right)^{-\kappa_i^*} \prod_{i=1}^n \frac{|M_i^*|^{1/2}}{|M_i|^{1/2}} \frac{\tau_i^{\kappa_i} \Gamma(\kappa_i^*)}{\Gamma(\kappa_i)}$$

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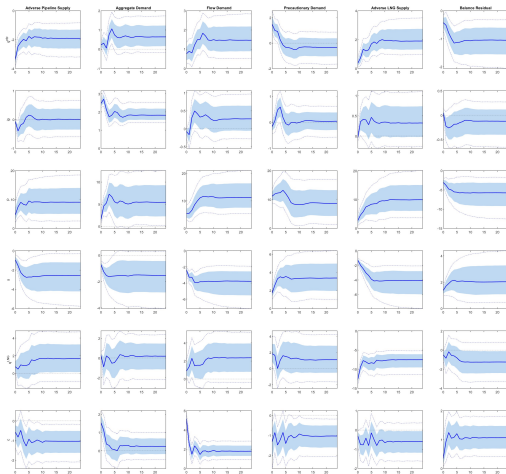
Identified structural shocks



Posterior median of each structural shock over the sample. The precautionary shock is serially uncorrelated ($\rho_1 = -0.01$), orthogonal to both supply shocks (correlations -0.06 and $+0.01$) and unpredictable from their lags; its largest realisations fall on the invasion month, the Hormuz closure, the pre-winter scrambles of 2021 and 2022 and the January 2020 demand collapse.

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Cumulative responses of all six variables to all six shocks



Rows: response variable. Columns: structural shock. Months 0–24; median with credible bands.

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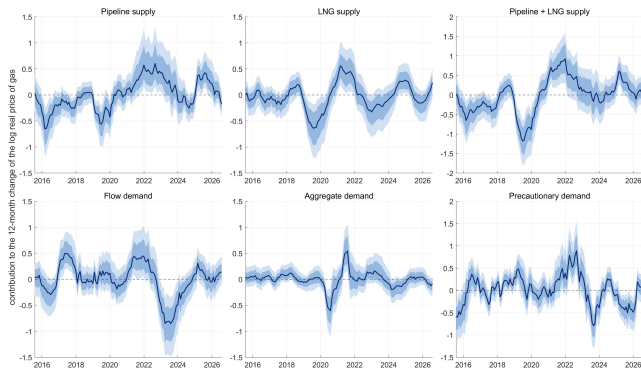
Realised variance of the price by regime

Shock	Full sample	2021–22 crisis	2026 crisis
Pipeline supply	14.2	11.8	10.3
Aggregate demand	8.2	2.2	1.1
Flow demand	13.9	11.5	3.6
Precautionary demand	46.2	59.4	71.9
LNG supply	8.3	3.8	5.9
Balance residual	4.3	3.4	3.6

Shares (%) of the sum of squared monthly price contributions within each window, computed draw by draw and reported at their posterior medians. Full sample 2014M8–2026M7; the 2021–22 window is September 2021 to December 2022; the 2026 window March to July 2026 (five months).

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Historical decomposition, all six shocks



Share of the price rise (%)

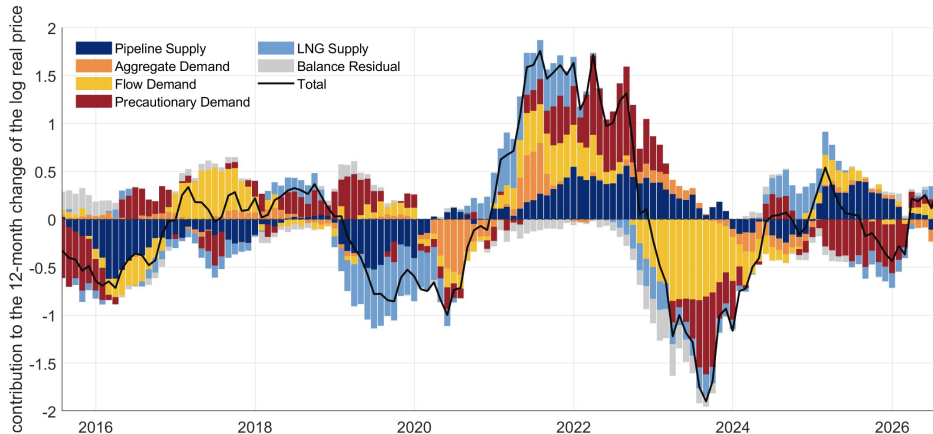
	2021–22	2026
Precautionary	69	64
Pipeline supply	43	–36
LNG supply	–12	52
Flow demand	3	2
Activity, residual	–4	20

Windows: Aug 2021 to the Aug 2022 peak (price $\times 4.2$); Feb to Jul 2026 ($\times 1.6$).

Contribution to the 12-month change of the log real price; median with 68% and 90% bands. Top: supply, with pipeline and LNG summed on the right. Bottom: demand.

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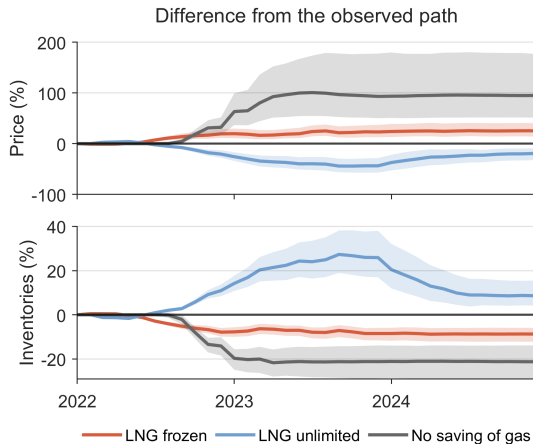
Historical decomposition, stacked



Contribution of each shock to the change of the log real price over the preceding twelve months, posterior medians stacked; black line = their sum.

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The three counterfactuals as gaps, with credible bands



Counterfactual minus observed (%); medians, 68% bands.

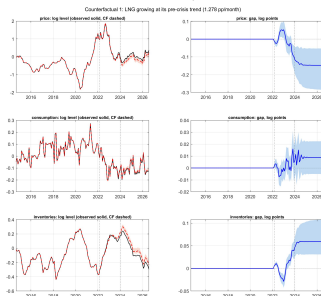
Gaps vs observed, December 2023 (%)

	price	stocks	cons.
LNG frozen	+23	−8	−2
LNG unlimited	−43	+26	+3
No saving of gas	+99	−21	+12

Unlimited LNG would have needed a doubling of monthly imports. From January 2024 LNG returns to its observed level, so that gap narrows (price −17% in July 2026). Of the saving of gas, about 30% is the mild winter and 70% autonomous.

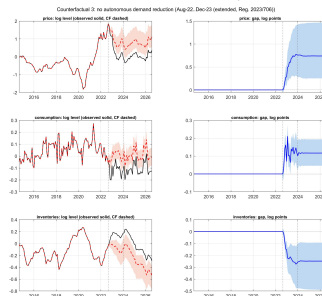
Two variants

LNG growing at its pre-crisis trend (1.28% per month)



In December 2023 the price is about 11% below the observed path: the realised expansion fell short of the pre-crisis trajectory, so the crisis-era interventions were necessary at the margin.

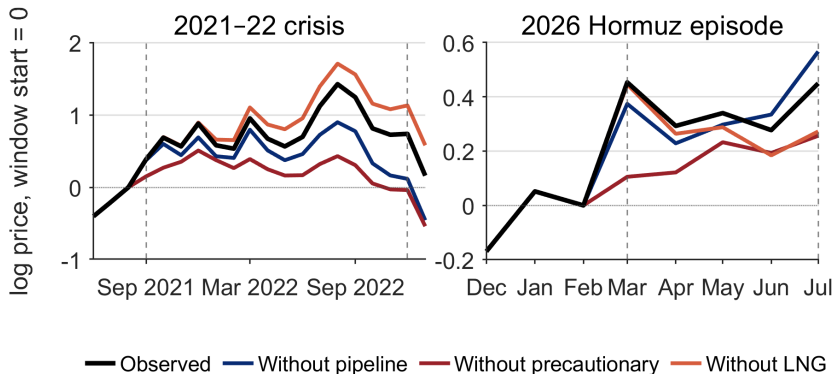
No saving of gas over the extended window (Aug 2022–Dec 2023)



Same pattern, larger magnitudes. The main text keeps the formal window of Regulation 2022/1369 because that is where the target was binding.

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The two crises: price paths

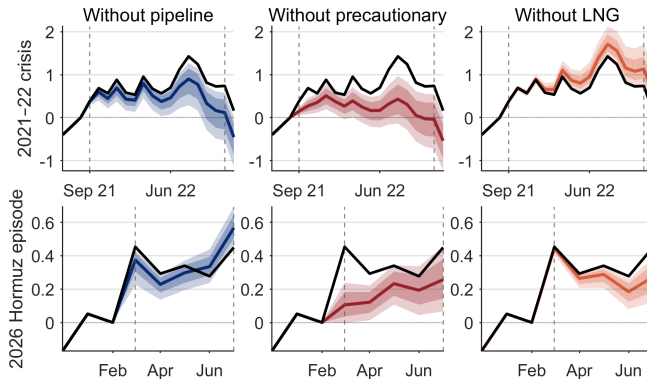


Observed log price (window start = 0) and the median paths without each channel's shocks. Dashed lines: crisis windows, September 2021 to December 2022 and March to July 2026.

[Credible bands](#)

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The crisis counterfactuals with credible bands



Observed log price (black) and the path without that channel's shocks: median, 68% and 90% bands. In 2026 the precautionary deletion separates at the March peak and the LNG deletion by July.

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Hormuz 2026, month by month

2026	Mar	Apr	May	Jun	Jul
Real price, % m/m	+57	-15	+5	-6	+19
LNG arrivals, % m/m	+3	-14	-2	-2	-16
Price without each channel's shocks since March, % vs observed					
Precautionary	-29	-16	-10	-8	-18
LNG	-1	-3	-5	-9	-16
Pipeline	-8	-6	-4	+6	+12
Recovered structural shocks					
Precautionary demand	+20.6	-11.9	-3.2	-1.1	+5.6
LNG supply	-3.9	-8.7	-3.9	-10.3	-25.6

Posterior medians. July 2026 quantities are a first release.

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Scenario design

Shock paths in units of the size of each draw's March 2026 shock (negative cargoes: fewer cargoes; positive precautionary: more hoarding); zero from January 2027.

	Aug	Sep	Oct	Nov	Dec
Cargoes (LNG supply)					
De-escalation	+0.25	+0.50	+0.25	0	0
Frozen conflict	-0.25	-0.25	-0.25	-0.25	0
Winter escalation	-0.75	-0.75	-0.50	-0.50	-0.25
Precautionary demand					
De-escalation	-0.50	-0.25	0	0	0
Frozen conflict	+0.25	+0.25	+0.25	+0.25	0
Winter escalation	+1.00	+1.00	+0.75	+0.50	+0.25

- *Largest gap over the no-new-shocks path: de-escalation $\times 0.72$ (December 2026), frozen conflict $\times 1.53$ and winter escalation $\times 4.27$ (January 2027).*
- *The March cargo shock was small, as arrivals still rose. Scaled to the March to July shocks, cargoes carry 4% to 22% of the winter-escalation peak.*
- *With no new shocks the price still rises 24% by July 2027: the model's own projection.*

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Robustness

Robustness runs on the paper's sample (data to May 2026, baseline share 47%).

- **Convergence.** A second chain of 2 million draws from a dispersed start reproduces every posterior median to the second decimal; effective sample sizes above 3,000 for β_{qp} , about 200 along the $\gamma_{qp}-\psi_1-\psi_2$ ridge.
- **Seasonal treatment.** Monthly dummies instead of X-13: twelve-month storage share 47% $\rightarrow$ 42%; β_{qp} and the weights unchanged.
- **Prior sensitivity.** Agnostic storage scales: 41%; demand prior loosened fivefold: 52% (α_{qp} moves toward smaller elasticities); re-weighting over a class of storage priors keeps the share in [40%, 50%]; dogmatic limit $\psi_1 = \psi_3 = 0$: 52%. Prior variation does not move the result.
- **Storage convention.** Stock change scaled by lagged supply (Rubaszek et al., Farag): 44% against 47%.
- **Composition of the precautionary shock.** Serially uncorrelated, orthogonal to current and future supply shocks; it records the expectational component of supply news, as in Kilian (2009) for the Gulf War.
- **Measurement.** With IEA supply data the share falls to 26%; we regard [26%, 47%] as the measurement band.

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Same identification, IEA Gas Trade Flows in place of JODI

Parameter	JODI baseline		IEA robustness	
	median	90% interval	median	90% interval
γ_{qp}	+0.116	[+0.052, +0.258]	+0.633	[+0.264, +1.170]
α_{yp}	-0.015	[-0.040, -0.002]	-0.010	[-0.030, -0.001]
α_{qy}	+0.695	[+0.407, +1.013]	+0.564	[+0.281, +0.831]
α_{qp}	-0.098	[-0.256, -0.014]	-0.032	[-0.097, -0.003]
ψ_1	-0.198	[-1.202, +0.071]	-0.613	[-0.970, -0.258]
ψ_2	-0.413	[-0.881, -0.231]	-0.264	[-0.603, -0.112]
ψ_3	+0.006	[-0.107, +0.087]	-0.033	[-0.149, +0.057]
β_{qp}	+0.160	[+0.032, +0.342]	+0.035	[+0.003, +0.110]
w_L	+0.228	[+0.192, +0.265]	+0.356	[+0.273, +0.446]
w_S	+1.797	[+1.614, +1.989]	+2.298	[+2.012, +2.657]
$\Pr(\beta_{qp} > \gamma_{qp})$	0.636		0.001	

Paper's sample (data to May 2026): the IEA series are not yet updated.

The IEA LNG series records scheduled terminal send-out, which cannot respond to a price within the month; JODI records cargo arrivals, which can. The LNG elasticity falls and the ordering of the two channels reverses: an elasticity attributed to a supply channel is a property of the measurement concept as much as of the market.

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